

True / False (40 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Within a twenty four hour period, a person could experience' is 'a globe's complete rotation'.
2. True or False: The correct answer to 'A hard-edge painting is most likely to be characterized by' is 'blurry color mixed on the painting's surface'.
3. True or False: The correct answer to 'What is the general relationship between temperature and entropy for diatomic gases?' is 'There is an inverse relationship, because at higher temperatures substances are more likely to be in a gaseous state.'.
4. True or False: The correct answer to 'The human stomach is protected from self-digestion by a layer of' is 'mucus.'.
5. True or False: The correct answer to 'A boy smacks another child in the face. The child cries because' is 'it was harmed'.
6. True or False: The correct answer to 'Joe and Mike both ran the samerace. Joe finished the race 4 minutes before Mike. If Mike finished the race at 4:02 p.m., what time did...' is '4:06 p.m.'.
7. True or False: The correct answer to 'Potential energy changes to kinetic energy when using a' is 'bicycle'.
8. True or False: The correct answer to 'If f is a linear transformation from the plane to the real numbers and if $f(1, 1) = 1$ and $f(-1, 0) = 2$, then $f(3, 5) =$ ' is '8'.
9. True or False: The correct answer to 'Which of the following policies best describes supply-side fiscal policy?' is 'Lower taxes on research and development of new technology'.
10. True or False: The correct answer to 'There is general consensus among paleoanthropologists that anatomically modern humans:' is 'migrated out of Africa into Southwest Asia between 150,000 and 100,000 years ago and then spread into Europe, Asia, Australia, and the Americas.'.
11. True or False: The correct answer to 'If a bird is moving through the sky, someone wanting to know the speed would' is 'watch slowly'.
12. True or False: The correct answer to 'Water evaporates less' is 'in a basement'.
13. True or False: The correct answer to 'Statement 1— Layer Normalization is used in the original ResNet paper, not Batch Normalization. Statement 2— DCGANs use self-attention to...' is 'True, True'.

14. True or False: The correct answer to 'According to the Law of Effect, behaviors followed by negative consequences' is 'occur less frequently'.
15. True or False: The correct answer to 'Which of the following brain structures is most centrally involved in hunger motivation?' is 'hypothalamus'.
16. True or False: The correct answer to 'Which list contains only nonliving things?' is 'coins, snakes, birds'.
17. True or False: The correct answer to 'Which of the following is true about a price floor?' is 'It will be located below the equilibrium price.'.
18. True or False: The correct answer to 'Find the remainder when 25^{1059} is divided by 23.' is '5'.
19. True or False: The correct answer to 'Staphylococcus aureus is...' is '...the term for the follicle after it has ruptured'.
20. True or False: The correct answer to 'Which of the following reflects the most light?' is 'snow'.

End of Paper